

Acyclic Graph Edges

Time limit: 1.00 s

Memory limit: 512 MB

Given an undirected graph, your task is to choose a direction for each edge so that the resulting directed graph is acyclic.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two distinct integers a and b : there is an edge between nodes a and b .

Output

Print m lines describing the directions of the edges. Each line has two integers a and b : there is an edge from node a to node b . You can print any valid solution.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
3 3 1 2 2 3 3 1	1 2 3 2 3 1